

HTI Methodology — Health Transformation Index

Audience: Data scientists, researchers, policymakers. **Version:** 4.7.0 | **Updated:** April 2026 **Route:** [/hti-dashboard](#)

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1. Overview

The **Health Transformation Index (HTI)** is a composite metric that tracks the overall state of U.S. healthcare transformation across six analytical pillars: Policy, Economics, Technology, Clinical, Equity, and Operations. It produces:

- A **national composite score** (0–100) updated quarterly
- **Six pillar sub-indexes** (0–100 each)
- **State Performance Index scores** for all 50 states across six pillar dimensions

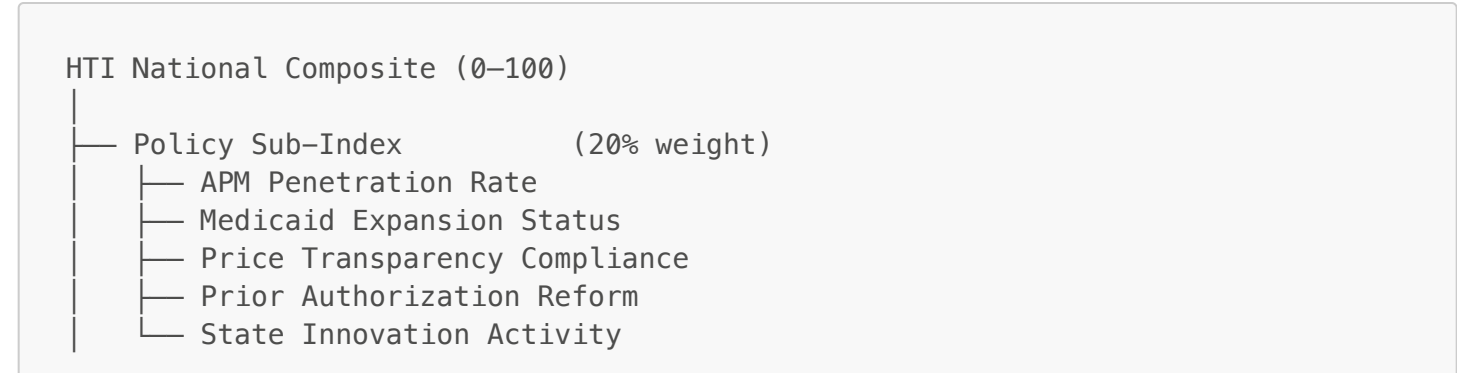
The HTI is designed to answer the question: *Is the U.S. healthcare system making progress toward transformation?* It is a macro-level tracking instrument, not an individual organization performance metric.

Score Interpretation

Score Range	Interpretation
85–100	Transformative leadership — sector is materially restructured
70–84	Advanced progress — significant structural change underway
55–69	Moderate progress — meaningful initiatives, incomplete adoption
40–54	Early stage — foundational activity with limited penetration
0–39	Pre-transformation — minimal structural change evident

Vermont's current national composite sits at **82/100** (Q1 2026), making it one of the highest-scoring states overall while facing acute financial sustainability challenges — a finding that illustrates why HTI measures transformation *infrastructure maturity*, not financial health.

2. Index Architecture



- Economics Sub-Index (25% weight)
 - VBC Contract Penetration
 - Hospital Operating Margin (inverted risk)
 - Payer Efficiency Ratio
 - Healthcare Spend Growth vs. GDP
 - APM Shared Savings Performance
- Operations Sub-Index (10% weight)
 - Administrative Overhead Rate
 - Revenue Cycle Friction Index
 - Workforce Operational Stability
- Technology Sub-Index (15% weight)
 - FHIR API Adoption Rate
 - EHR Interoperability Score
 - Telehealth Penetration
 - AI Clinical Tool Deployment
 - Health IT Investment per Beneficiary
- Clinical Sub-Index (20% weight)
 - Quality Composite (HEDIS/Stars)
 - Preventable Hospitalization Rate
 - Hospital Readmission Rate
 - Patient Experience Score
 - Workforce Sufficiency Index
- Equity Sub-Index (10% weight)
 - Racial Disparity Index
 - Rural Access Score
 - SDOH Investment Rate
 - Uninsured Rate by Subgroup
 - Community Health Worker Penetration

3. Data Sources

Federal Data Sources

Source	Data Elements	Update Frequency
CMS Quality Payment Program (QPP)	APM participation, MIPS scores	Annual
CMS Innovation Center (CMMI)	CMMI model participation, shared savings	Annual
CMS Medicare Fee-for-Service	Quality measures, utilization	Quarterly
CMS Medicaid.gov	State Medicaid expansion, enrollment	Monthly
AHRQ HCUP	Hospital utilization, preventable admissions	Annual
ONC Health IT Dashboard	EHR adoption, interoperability measures	Annual
HRSA	HPSA designations, primary care workforce	Annual
CDC BRFSS	Behavioral risk factors, preventive care	Annual
Census Bureau ACS	Income, insurance, SDOH demographics	Annual
KFF State Health Facts	Insurance coverage, policy tracking	Monthly

Proprietary HTR Scoring

Some components are scored using HTR's proprietary assessment methodology rather than a single public data source:

- **State Innovation Activity** — HTR analysts score each state's legislative and regulatory innovation portfolio on a 0–10 scale, updated quarterly
- **Payer Innovation Leadership** — scored annually based on APM contracts disclosed, digital health programs, and equity reporting
- **Health IT Investment** — based on CMS innovation funding, ONC grants, and disclosed health system IT budgets

Data Lag

Most federal data sources have 12–18 month reporting lags. HTR applies a **currency adjustment** for lagged data: metrics from the most recent available year are weighted at 100% of face value; data from two years prior is weighted at 85%; three years prior at 70%. This prevents the index from being dominated by stale data in high-lag components.

4. Pillar Sub-Indexes

Policy Sub-Index (20% weight)

Tracks legislative and regulatory transformation across five components:

APM Penetration Rate (30% of pillar) Percentage of total healthcare spending flowing through alternative payment models (two-sided risk or higher). Source: CMMI annual participation data + KFF.

Scale: 0–100, where 100 = ≥60% of spending in two-sided risk APMs.

Medicaid Expansion Status (20% of pillar) Binary (0 or 100) for whether the state has adopted ACA Medicaid expansion, plus a partial credit score for states with partial expansion or active negotiations.

Price Transparency Compliance (20% of pillar) CMS tracks hospital compliance with the Hospital Price Transparency Final Rule. Scored 0–100 based on share of hospitals with compliant machine-readable files.

Prior Authorization Reform (15% of pillar) Scored 0–100 based on: state PA reform legislation, CMS Interoperability and PA Final Rule compliance, average PA approval time.

State Innovation Activity (15% of pillar) HTR proprietary score assessing: active 1115 waivers, SIM grants, global budget programs, all-payer model participation, reinsurance programs.

Economics Sub-Index (25% weight)

VBC Contract Penetration (35% of pillar) Share of commercially insured and Medicaid beneficiaries in value-based payment arrangements (any tier, including P4P). Source: HCPLAN annual survey.

Hospital Operating Margin Stability (20% of pillar) Inverse score — higher scores indicate more stable margins. Measured as percentage of hospitals with operating margin ≥ -2%. Source: CMS Cost Reports.

Healthcare Spend Growth vs. GDP (25% of pillar) Ratio of healthcare expenditure growth to GDP growth. Score of 100 when healthcare grows ≤ GDP; 0 when growing at 2× GDP rate or higher. Source: CMS National Health Expenditure Accounts.

APM Shared Savings Performance (20% of pillar) Net shared savings generated by CMS Innovation models as a percentage of benchmarked spend. Source: CMMI evaluation reports.

Operations Sub-Index (10% weight)

Tracks the administrative and operational execution capacity of the U.S. healthcare system — the infrastructure layer that determines whether policy and clinical reforms can actually be carried out.

Administrative Overhead Rate (35% of pillar) National administrative expenditures as a percentage of total healthcare spending, benchmarked against the prior-year baseline and international comparators. Lower overhead relative to prior periods scores higher. Source: CMS National Health Expenditure Accounts, Commonwealth Fund international comparisons.

Revenue Cycle Friction Index (35% of pillar) National aggregate of clean claim submission rates, average days in accounts receivable, and denial rates across Medicare and Medicaid claims. Lower friction scores higher. Source: CMS cost reports, HFMA benchmark surveys.

Workforce Operational Stability (30% of pillar) National composite of healthcare workforce turnover rates, clinical vacancy rates, and credentialing cycle times across key workforce categories. Higher stability scores higher. Source: HRSA workforce data, ACHE annual operational surveys, BLS Occupational Employment Statistics.

Technology Sub-Index (15% weight)

FHIR API Adoption (30% of pillar) Share of hospitals and clinics with certified FHIR R4 Patient Access and Provider Access APIs. Source: ONC Health IT Dashboard.

EHR Interoperability Score (25% of pillar) Share of clinicians who send, receive, find, and integrate patient health information from outside their organization (ONC four-part interoperability). Source: AHA IT Supplement.

Telehealth Penetration (25% of pillar) Share of eligible outpatient encounters delivered via telehealth (audio-visual). Source: CMS telehealth utilization data.

AI Clinical Tool Deployment (20% of pillar) HTR proprietary score based on disclosed AI deployments by major health systems, FDA AI/ML-enabled device approvals, and survey data on clinical AI use.

Clinical Sub-Index (20% weight)

Quality Composite (35% of pillar) Weighted average of HEDIS commercial, Medicaid, and Medicare measures at national level. Source: NCQA Quality Compass.

Preventable Hospitalization Rate (25% of pillar) AHRQ Prevention Quality Indicators (PQI) composite, inverted (lower PQI = higher score). Source: HCUP.

Hospital Readmission Rate (20% of pillar) CMS hospital readmission reduction program composite. Source: CMS.

Workforce Sufficiency Index (20% of pillar) Composite of primary care physician supply per 100,000 population and nursing vacancy rates. Source: HRSA, AHA Annual Survey.

Equity Sub-Index (15% weight)

Racial Disparity Index (35% of pillar) Composite gap score across five clinical measures comparing non-Hispanic white to Black and Hispanic populations. Source: NCQA HEQ measures, AHRQ National Healthcare Quality and Disparities Report.

Rural Access Score (25% of pillar) Share of rural population within 30 minutes drive of primary care, emergency care, and OB/GYN services. Source: HRSA, USDA Rural-Urban Continuum.

SDOH Investment Rate (20% of pillar) Per-capita social services spending relative to healthcare spending. Source: NYU Stern Global Center for Sustainable Business; state budget data.

Uninsured Rate by Subgroup (20% of pillar) Focus on non-elderly, low-income, and minority uninsured rates. Scored 0–100 inverted (lower uninsured = higher score). Source: ACS.

5. Composite Score Construction

Weighting

Pillar	Weight	Rationale
Policy	20%	Sets the rules; slower-moving than other pillars
Economics	25%	Financial incentives are the most powerful behavioral lever
Operations	10%	Execution capacity; introduced at baseline weight as national data infrastructure matures
Technology	15%	Enabler of all other transformation; reduced from 20% as Operations introduced
Clinical	20%	Ultimate purpose — actual care delivery improvement
Equity	10%	Critical cross-cutting lens; reduced from 15% pending improved national SDOH and algorithmic bias data coverage; scheduled for review in the 2027 methodology revision

Weights are reviewed annually by HTR's methodology committee and may be adjusted as the transformation landscape evolves.

Normalization

Each component is normalized to a 0–100 scale before pillar aggregation:

$$\text{Normalized Score} = ((\text{Raw Value} - \text{Min Benchmark}) / (\text{Max Benchmark} - \text{Min Benchmark})) \times 100$$

Where:

- Min Benchmark** = the value that represents pre-transformation baseline (score of 0)
- Max Benchmark** = the value that represents leading-edge transformation (score of 100)

Benchmarks are set by HTR's methodology committee based on literature, CMS targets, and observed leading-edge state performance.

Composite Calculation

$$\text{HTI} = (\text{Policy} \times 0.20) + (\text{Economics} \times 0.25) + (\text{Operations} \times 0.10) + (\text{Technology} \times 0.15) + (\text{Clinical} \times 0.20) + (\text{Equity} \times 0.10)$$

Trend Arrow Logic

The HTI Dashboard displays trend arrows (▲ ▼ —) based on quarter-over-quarter change:

Change	Arrow
+1.5 or more	▲ (improving)
-1.5 or less	▼ (declining)

Change	Arrow
Between -1.5 and +1.5	— (stable)

6. State Performance Index

Each state receives a Performance Index (0–100) across six dimensions, displayed in the State Dashboard.

Six Dimensions

Dimension	Weight	Key Metrics
Performance Score	Composite	Weighted average of all six dimensions below
Cost Index	—	Total healthcare spend per capita relative to national median (inverted)
Quality Score	—	HEDIS + Stars composite for commercial and Medicaid populations
Access Score	—	Uninsured rate, primary care supply, telehealth adoption
Equity Score	—	Racial disparity index + rural access + uninsured by subgroup
Innovation Score	—	APM penetration + VBC contracts + CMMI model participation

State Data Table (Sample)

State	Performance Score	Cost Index	Quality Score	Access Score	Equity Score	Innovation Score
Vermont	82	1.18	78	73	81	85
Massachusetts	79	1.24	82	77	74	84
Minnesota	76	1.09	80	71	71	79
Mississippi	31	0.89	42	28	22	29

Note: The Cost Index is not inverted in the raw data — a cost index of 1.18 means 18% above national median. The State Dashboard inverts this for scoring purposes (lower cost = higher performance).

7. Update Cadence

Component	Update Frequency	Next Update
HTI National Composite	Quarterly	Q2 2026 (June)
Pillar Sub-Indexes	Quarterly	Q2 2026 (June)
State Performance Index	Semi-annual	Q3 2026 (September)
HTI Dashboard indicators	Real-time where available	Continuous
Methodology review	Annual	January 2027

8. Limitations & Interpretive Guidance

What the HTI Is Not

- **Not a financial health indicator.** Vermont's high HTI score coexists with severe hospital financial distress — the index measures transformation infrastructure, not financial sustainability.
- **Not a quality ranking.** The HTI captures adoption of transformation mechanisms, not ultimate health outcomes. A state can score high by adopting APMs that have not yet improved outcomes.
- **Not an individual organization assessment.** The HTI applies to states and the national system. It should not be used to evaluate individual health systems, payers, or providers.

Data Quality Considerations

- Federal data sources have inconsistent reporting requirements and voluntary vs. mandatory disclosure
- CMMI model participation data is precise; health system AI adoption data is based on disclosure and surveys
- Rural health data is less reliable than urban data due to small sample sizes in surveys
- Equity data is systematically less complete for smaller racial/ethnic subgroups

Geographic Aggregation

State-level scores mask significant within-state variation. Vermont's urban-rural divide in access scores is larger than the state average suggests. Users analyzing specific geographies should supplement state scores with county-level data from AHRQ, CDC, and HRSA.

9. HTI Dashboard (Platform)

The HTI Dashboard at [/hti-dashboard](#) displays:

- **National HTI Score** — large composite display with trend arrow
- **System Vitals Ticker** — scrolling real-time indicators (APM penetration, VBC spend share, etc.)
- **Pillar Cards** — six cards showing each pillar sub-index with trend arrow and top-line summary
- **Methodology Link** — links to this document
- **Historical Trend** — quarterly HTI history chart

The dashboard is publicly accessible (no login required). All indicators reflect the most recent quarterly release.